ECON 101: Macroeconomics
Solutions – Quiz 6

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Part A — Multiple Choice Questions (1 mark each)

1. The Bank of Canada's primary monetary policy instrument is:
Correct answer: (C) The target overnight interest rate.

Explanation:

- The Bank of Canada implements monetary policy by setting a **target for the overnight interest rate** (key policy rate).
- Open-market operations and settlement balances are used *to keep* the overnight rate near this target.
- The required reserve ratio is not the main instrument in Canada.

2. If the Bank of Canada lowers the target overnight rate, the most immediate effect is:
Correct answer: (D) Increase in settlement balances and lower short-term rates.

Explanation:

- To push the overnight rate *down*, the Bank of Canada typically **adds liquidity** to the system (e.g., via open-market purchases or higher settlement balances).
- More settlement balances increase the supply of overnight funds, lowering the overnight rate and other short-term market rates.
- So the immediate effect is more settlement balances and lower short-term interest rates.

3. An open-market *sale* of government securities by the Bank of Canada will:
Correct answer: (B) Reduce bank reserves and raise interest rates.

Explanation:

- When the Bank *sells* government bonds, banks and dealers pay by giving up reserves/settlement balances.
- This **reduces the supply** of reserves in the system.
- With fewer reserves, the overnight rate and other interest rates tend to rise.
- Hence it is a contractionary move.

4. The monetary policy transmission mechanism begins with:
Correct answer: (C) A change in the policy interest rate.

Explanation:

- The first step is a change in the **target overnight rate** (policy interest rate).
- This then affects other interest rates, the exchange rate, asset prices, and eventually spending components (C, I, NX).
- So the mechanism begins with the Bank's policy rate, not directly with AD or LRAS.

5. If the Bank of Canada raises interest rates, the Canadian dollar tends to:
Correct answer: (B) Appreciate, reducing net exports.

Explanation:

- Higher Canadian interest rates make Canadian financial assets more attractive to foreign investors.
- Capital inflows increase demand for the Canadian dollar.
- The dollar **appreciates**, making exports more expensive and imports cheaper.
- Net exports (NX) decline.

6. Which of the following is part of the Bank of Canada's inflation-control strategy?
Correct answer: (B) A 2% inflation target midpoint.

Explanation:

- The Bank of Canada and the federal government agree on an inflation-control target range, with a **2% midpoint** (typically 1%–3% band).
- The Bank does not target the money supply or exchange rate directly, and it does not impose wage ceilings.

7. Contractionary monetary policy in the AD–AS model will:
Correct answer: (C) Shift AD to the left and lower real GDP in the short run.

Explanation:

- Higher interest rates reduce C, I, and NX, which lowers aggregate demand.
- In the AD–AS diagram, this appears as a **leftward shift of AD**.
- In the short run, with upward-sloping SRAS, real GDP falls and the price level tends to fall or rise more slowly.

Part B – Problem (13 marks)

Problem 8 — Taylor Rule and Transmission Mechanism

We are given the reaction function:

$$i = 1 + 1.2(\pi - 2) + 0.4(Y - Y^*),$$

where:

- i = target overnight rate (percent),
- π = actual inflation rate (percent),
- 2 = inflation target midpoint (percent),
- $Y - Y^*$ = output gap (percent).

Data:

$$\pi = 3.1\%, \qquad Y - Y^* = +0.8\%.$$

(a) Compute the recommended Bank of Canada target overnight rate (4 marks)

Step 1: Compute the inflation deviation from target.

$$\pi - 2 = 3.1 - 2.0 = 1.1 \quad \text{(percentage points above target).}$$

Step 2: Compute the inflation term $1.2(\pi - 2)$.

$$1.2 \times 1.1 = 1.32.$$

Step 3: Compute the output gap term $0.4(Y - Y^*)$.

$$Y - Y^* = +0.8 \Rightarrow 0.4 \times 0.8 = 0.32.$$

Step 4: Plug into the reaction function.

$$i = 1 + 1.2(\pi - 2) + 0.4(Y - Y^*) = 1 + 1.32 + 0.32 = 2.64.$$

Answer:

$$\boxed{i = 2.64\% \text{ (target overnight interest rate)}}$$

(b) Is the resulting policy expansionary or contractionary? Explain. (2 marks)

Step 1: Interpretation of the rule.

- Inflation is above target (3.1% vs 2%).
- The output gap is positive (economy operating 0.8% above potential).

Step 2: Policy stance.

- The rule recommends a relatively *higher* interest rate (2.64%) to cool an overheating economy.
- Compared to a neutral or lower rate, this implies tighter monetary conditions.

Conclusion:

$$\boxed{\text{The implied stance is contractionary (tightening), intended to reduce inflationary pressure.}}$$

(c) Transmission mechanism: effect of a higher policy rate (4 marks)

Step-by-step linkage:

Step 1: From policy rate to market interest rates.

- The Bank of Canada raises the target overnight rate to 2.64%.
- Short-term money market rates, bank prime rates, and other lending rates rise.
- Longer-term interest rates may also rise as expectations adjust.

Step 2: Effect on consumption.

- Higher borrowing costs make credit-financed purchases (cars, appliances, renovations, housing) more expensive.
- Some households delay or cancel spending plans.
- Consumption spending falls, especially on interest-sensitive goods.

Step 3: Effect on investment.

- Firms face a higher cost of financing new capital projects.
- Some investment projects that were profitable at lower rates become unprofitable.
- Business investment in machinery, buildings, and technology falls.

Step 4: Effect on exchange rate and net exports.

- Higher Canadian interest rates attract foreign capital (seeking higher returns).
- Demand for the Canadian dollar rises, causing the dollar to **appreciate**.
- Canadian exports become more expensive in foreign currency, and imports become cheaper for Canadians.
- Net exports (NX) decrease.

Step 5: Effect on aggregate demand.

- Lower consumption (C), lower investment (I), and lower net exports (NX) all reduce total planned spending.
- In the AD–AS framework, this is represented by a **leftward shift in the AD curve**.

(d) AD–AS explanation: bringing inflation back toward 2% (3 marks)

Initial situation:

- Inflation is above target (3.1%) and the output gap is positive ($Y > Y^*$).
- In the AD–AS diagram, the economy is at an equilibrium where AD intersects SRAS to the *right* of LRAS.

Effect of contractionary monetary policy:

- Higher interest rates reduce C, I, and NX, shifting the AD curve left (from AD_0 to AD_1).
- In the short run, this moves the equilibrium closer to the vertical LRAS line at Y^*.
- Real GDP falls toward potential, and the price level (and inflation rate) begin to decline.

Medium-run adjustment:

- With the output gap shrinking or closing, upward pressure on wages and prices diminishes.
- Inflation gradually moves back toward the 2% target midpoint.

Summary:

- Contractionary policy: AD shifts left.
- Short run: lower real GDP and lower inflation (or lower inflation growth).
- Medium run: economy returns to potential GDP with inflation near the 2% target.

End of Solutions